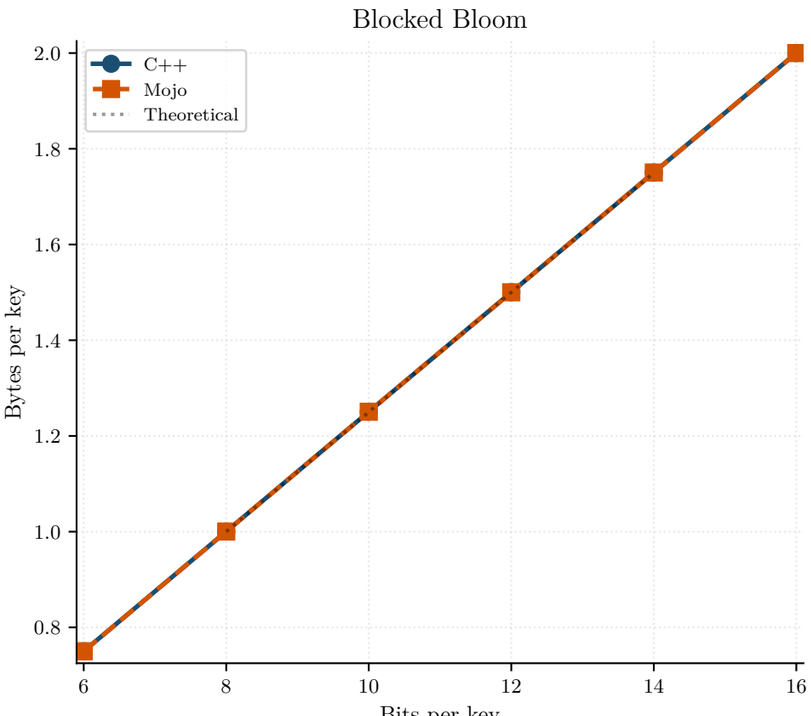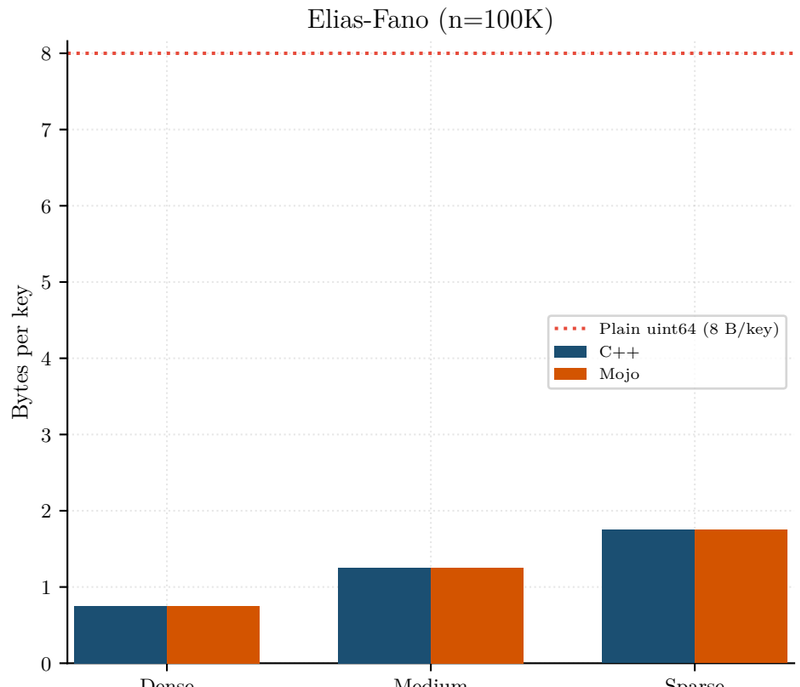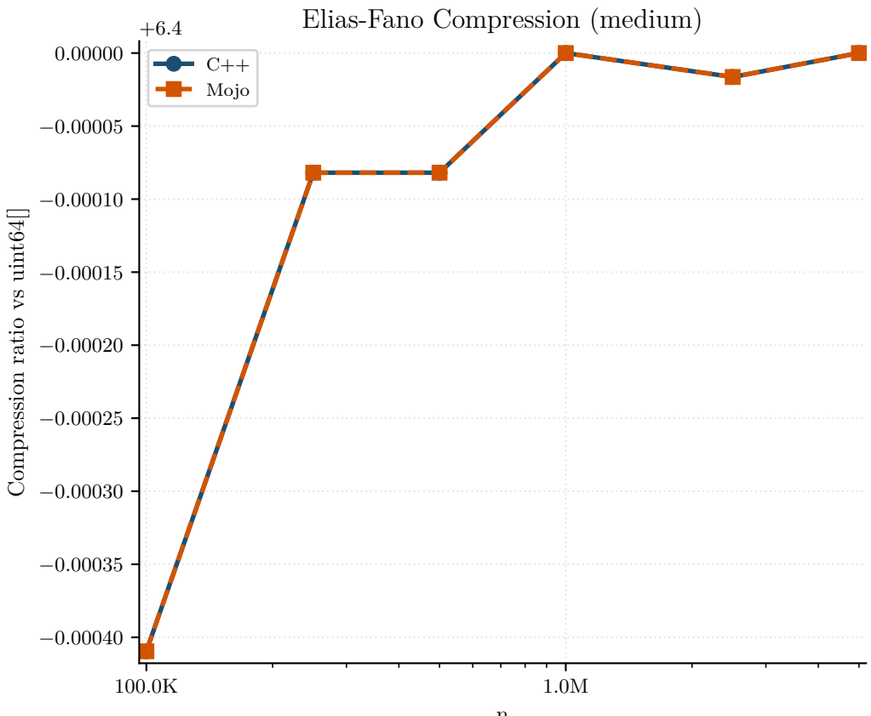Memory Efficiency Analysis